

Lecture 1: Introduction to fermentation

The core of microbial fermentation is using the wide diversity of microorganisms available.

1.1 Straightforward approach

This approach utilizes the natural processes of microorganisms for product formation. There's minimal intervention, except for slight changes in the environment, nutrient supply, or substrate composition to optimize production.

This is effective when:

- The natural microorganism produces the product at a good rate.
- It's scalable and economically viable.

Examples could include yeast used in ethanol production or bacteria used for citric acid fermentation.

1.2 Cell factory approach

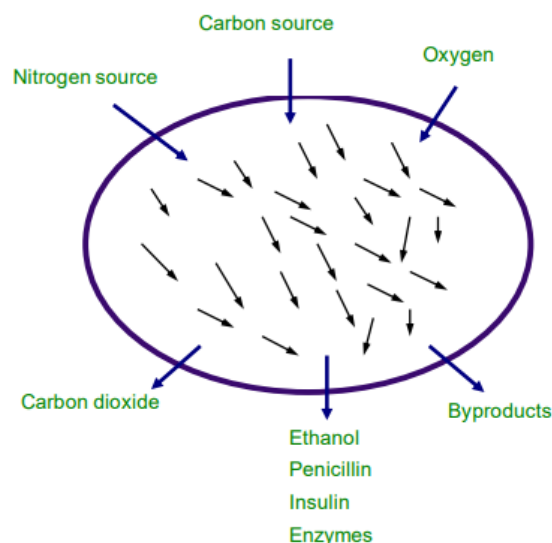
Focuses on transferring a desired metabolic pathway from a naturally occurring organism into a microorganism that is easier to handle in industrial settings (e.g., bacteria or yeast that are easier to grow)

Especially useful when the natural producer is slow-growing, difficult to cultivate, or not well understood.

Common in pharmaceutical production (e.g., insulin produced by genetically modified *E. coli* or yeast).

Steps:

1. Identify the product and corresponding metabolic pathway.
2. Transfer the pathway into scalable microorganisms
3. Optimize the cultivation conditions in the new host.



1.3 Choosing between approaches

Straightforward approach	Cell factory approach
The organism is easy to grow	The original organism grows too slowly
The production process is already well understood	There is little knowledge of its biology
Scaling and cost-efficiency are possible with minimal modifications.	A faster, more adaptable microorganism can be used to produce the product efficiently.

1.4 Optimization of culture conditions

Once the right microorganism is selected, production can be maximized by optimizing:

- **Process variables:** temperature, pH, oxygen levels, agitation, etc.
- **Medium composition:** Selecting the correct balance of nutrients (carbon, nitrogen, trace elements) to support growth and product formation.
- Adjusting these conditions improves productivity, yield, and cell health.

1.5 Strain improvement

Strain improvement strategies involve:

- **Genetic modification** to increase the production of the target product.
- **Eliminating by-products** that waste resources or slow down the production of the desired product.
- **Risks:** Deleting by-products could inadvertently harm the organism's survival, which is why strain improvement is more successful with well-known organisms.

1.6 Scale-Up challenges

Scaling up from lab-scale bioreactors (1L) to industrial-scale bioreactors (several thousand liters) introduces challenges:

- **Non-homogeneous conditions:** Large bioreactors may have uneven oxygen, nutrient, and temperature gradients.
- Cells experience **stress** from fluctuating conditions (e.g., oxygen starvation at the bottom of the tank and nutrient depletion at the top).
- Scale-up solutions include better mixing, aeration strategies, and optimized reactor designs.

1.7 Factory characteristics

Microorganisms optimized for industrial use in cell factories should ideally have the following characteristics:

- **Efficient substrate utilization:** Ability to convert available substrates into products with minimal waste.
- **Wide substrate range:** Flexibility to grow on a variety of carbon and nutrient sources.
- **High yield and productivity:** Maximized product formation from minimal substrate input and high production rates.
- **Minimized by-product formation:** Reduces waste and increases efficiency.
- **Morphology suited for bioreactor processes:** Organisms should be easy to handle in industrial reactors (unicellular organisms are often preferred over filamentous ones for ease of control).
- **Amenable to genetic modification:** Easy to manipulate genetically for improvement.

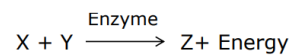
Properties	Bacteria	Yeast	Fungi
Easy for genetical modifications	+	+	±
Easy to cultivate	+	+	±
Fast growth	+	+	±
Ability to grow on complex substrates	-	±	+
Enzymes producer	±	±	+
Post translational modifications	-	±	+
Secondary metabolites producers	-	-	+

Lecture 2: Cell metabolism

- **Biochemistry** refers to the potential reactions and pathways that can take place in an organism. For example, humans don't have the biochemical pathways to produce ethanol, but yeast does.
- **Physiology**, focuses on which reactions and pathways are actually active under certain conditions. For instance, two *Saccharomyces* cells can have identical biochemistry, but under different conditions (one anaerobic, one aerobic), they utilize different pathways.

Metabolism includes both **catabolism** (breaking down molecules for energy) and **anabolism** (building up molecules for growth). Catabolism releases energy (exothermic) while anabolism consumes energy (endothermic). Together, these processes maintain the energy balance of the cell.

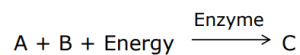
➤ Exothermic processes



CATABOLISM

- Degradation of carbohydrates to give CO₂ and water
Building blocks + reducing power + energy

➤ Endothermic processes



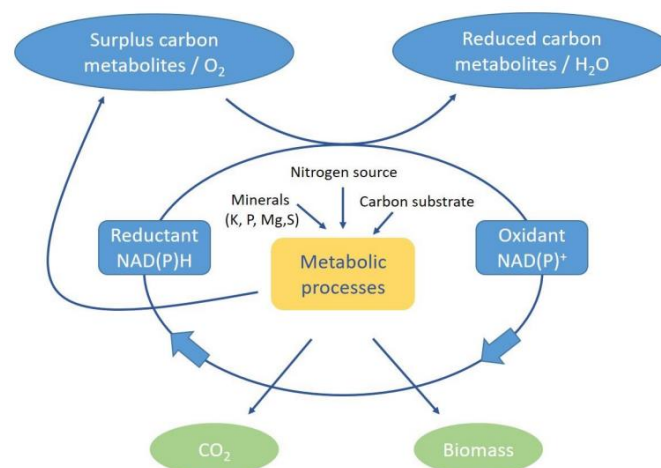
ANABOLISM

- Building up of cell materials and their intermediate precursors

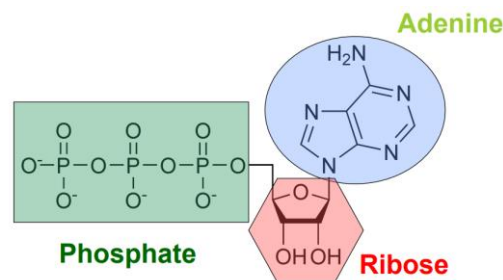
CATABOLISM + ANABOLISM = METABOLISM

2.1 Electron and Energy carriers

NAD(P)H and **NAD(P)⁺** are electron carriers, playing essential roles in cellular redox reactions. NAD(P)⁺ is primarily used in catabolism, while NAD(P)H is used in anabolic processes. These are mainly derived from vitamins and even amino acids through a less efficient path.



ATP is the main energy currency of the cell. Through processes like **oxidative phosphorylation** and **substrate-level phosphorylation**, the cell transfers energy from nutrients to ATP, which powers cellular functions.



2.2 Catabolic pathways

2.2.1 Embden-Meyerhof-Parnas (EMP) Pathway (Glycolysis)

This pathway breaks down glucose into pyruvate, producing ATP and NADH. It's active in both aerobic and anaerobic conditions and provides five of the nine principal precursors for cellular biosynthesis.

2.2.2 Pentose Phosphate (PP) Pathway

This pathway plays a critical role in the production of NADPH (necessary for biosynthetic reactions) and pentoses (used in nucleic acid synthesis). Cells in the exponential growth phase prioritize the PP pathway due to their high demand for nucleotides and reducing power.

When cells are in the exponential phase, the ratio for the pentose phosphate (PP) pathway is 2:1. However, when food becomes scarce, the ratio shifts to 1:20 in favor of the Embden-Meyerhof pathway (glycolysis) because cells no longer need to replicate, but they do need to survive. This shift is ideal for glycolysis.

2.2.3 Krebs cycle (TCA cycle)

The Krebs cycle completes the oxidation of acetyl-CoA, producing energy and essential intermediates for biosynthetic processes. It's a key energy generator in aerobic metabolism but also supplies precursors for other pathways. However, there's always a trade-off between energy production and intermediate generation so if it focuses on energy production over precursor supplying, the amount of the second one will decrease.

2.2.4 Glyoxylate bypass

Anaplerotic reactions do not completely replace metabolic pathways but help replenish intermediates. The **glyoxylate bypass**, a shortcut within the TCA cycle that allows the cell to bypass the decarboxylation steps, conserving carbon for biosynthesis. This bypass is especially useful when cells are growing on carbon sources like **acetate**, where carbon conservation is crucial. It enables the cell to generate key precursors such as malate and succinate more efficiently, supporting biosynthetic processes rather than just energy production.

2.2.5 Gluconeogenesis

Gluconeogenesis is essentially glycolysis upside down, as most of the enzymes are reversible and can work both ways. However, there are two key points where the direction is controlled by the cell. Phosphofructokinase is one of the most tightly regulated enzymes in the cell, and its bypass, along with the bypass of pyruvate kinase, is crucial for gluconeogenesis to occur."

2.3 Aerobic and anaerobic metabolism

In **aerobic** conditions, the electron transport chain uses oxygen as the final electron acceptor. This process is highly efficient, producing up to 38 ATP molecules per glucose (in theory). The main contributor to energy production is **oxidative phosphorylation**, where electrons are transferred through a series of carriers, generating a proton gradient used to produce ATP.

In **anaerobic** conditions, cells switch to fermentation, a less efficient process. In this scenario, NADH accumulates, and cells must rely on **substrate-level phosphorylation** to generate ATP. This results in the production of various fermentation byproducts (e.g., lactic acid or ethanol), which can become toxic at high concentrations. To maintain energy levels comparable to aerobic conditions, more substrate is needed, which increases the accumulation of unwanted byproducts.

2.4 Metabolite production

2.4.1 Primary metabolites

Primary metabolites, such as ethanol or lactic acid, are directly involved in cellular growth and energy production. These metabolites are primarily produced during the exponential growth phase when cells are rapidly dividing.

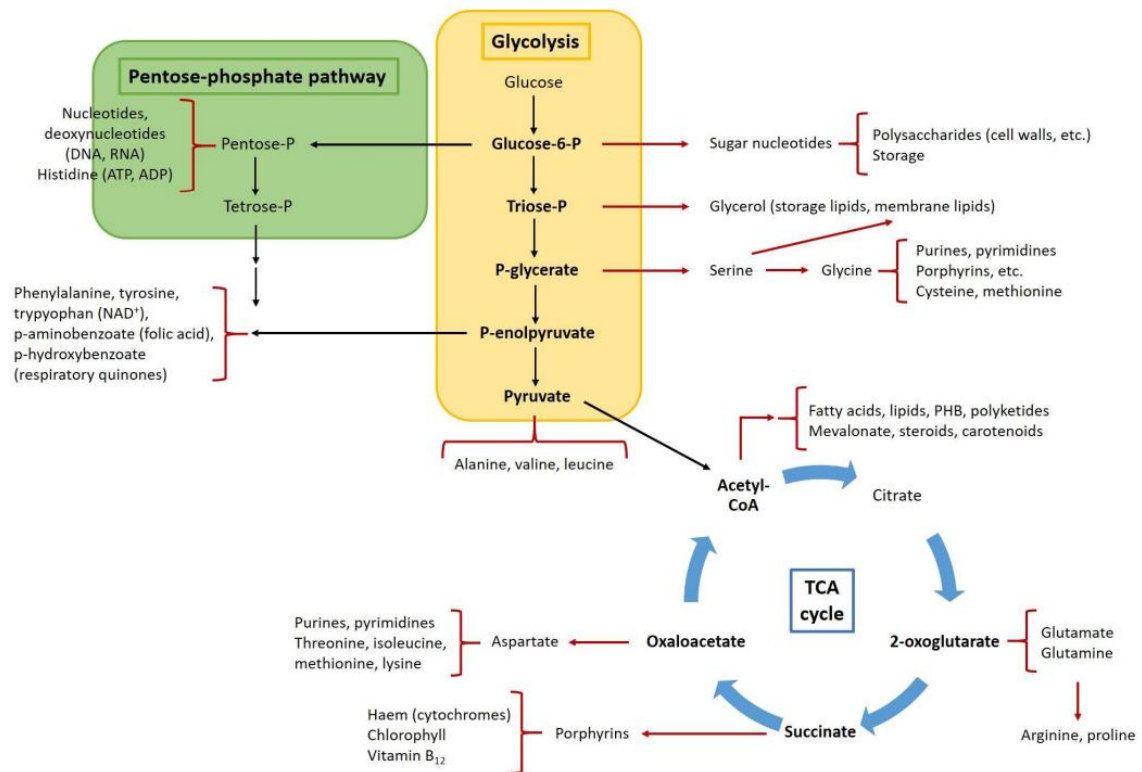
2.4.2 Secondary metabolites

Secondary metabolites, such as antibiotics or pigments, are not essential for growth or energy production. Their production typically occurs during the stationary phase when cell division slows or halts. However, their production can also be influenced by manipulating growth conditions. For example, to maximize secondary metabolite accumulation (like penicillin), cells may be grown under anaerobic conditions, even though the metabolite itself may not be a product of anaerobic metabolism. Secondary metabolites can arise from the accumulation of intermediates from primary metabolism, and forcing cells to grow slowly can trick them into entering a stationary-like phase, promoting secondary metabolite production.

Not all microorganisms can produce secondary metabolites, but filamentous fungi are particularly efficient at generating them. These organisms are highly effective at producing secondary metabolites under nutrient-limiting conditions, whether in aerobic or anaerobic environments. Anaerobic conditions tend to enhance the accumulation of certain primary metabolites, providing more precursors for secondary metabolite synthesis.

2.5 Microbial yield

The overall growth yield of a microorganism depends on various factors, including the carbon source, the metabolic pathway used, and the efficiency of ATP production. Aerobic conditions generally result in higher yields due to more efficient energy generation through oxidative phosphorylation. Anaerobic conditions, while necessary for producing certain products, tend to reduce yields due to the lower efficiency of fermentation pathways.



Lecture 3: Fermentation media design

3.1 Cellular composition and minimum requirements

The dry mass of a cell is composed of:

- Proteins, RNA, lipids, polysaccharides, DNA, metabolites and minerals (ash when burned).
- Amino acids, nucleotides, fatty acids, glycerol, sugars.

For media design, it's important to know the dry matter composition of the cell.

The **minimum requirements** for microorganisms are:

- Water, Carbon source, Nitrogen source and oxygen (if aerobic)
- When containing minerals and vitamins, it's a complex media.

3.3 Criteria for nutrient selection in large scale production

For large scale production, media needs to fulfill the following criteria.

- Maximum yield of product per gram of substrate, converting substrate into product.
- Maximum concentration of product, higher product concentration simplifies processing.
- Maximum rate of product formation, the speed at which the product is formed during the fermentation process.
- Minimum yield of undesired products
- Consistent quality of process along time
- Minimal problems during media preparation and sterilization. Not all media can be sterilized, some may require alternative sterilization methods (e.g., filtration) if they are heat-sensitive.
- Minimal problems in other aspects of the production (aeration, agitation, extraction and purification, ...)

3.4 Media

Complex	Defined
Nutrients in complex form, not well defined	Defined nutrients, might be necessary to add several components
Batch variations, irreproducible results	Well defined process
Problems in sterilization, precipitation	Less problems in sterilization
Color and smell	Clear solutions
Complicates product recovery and effluent treatment	Gives a simple and well defined downstream processing
Higher yields and productivities	Complex components must be synthesized by the cell
Easy medium design	Hard media design
Complicated process monitoring and control	Simplifies monitoring and control
Low price	Expensive

Complex media tends to be faster for industrial fermentation since cells don't have to bio-synthesize that many compounds. Defined media is better for research purposes since we know the exact composition.

Exam relevant:

- **Minimal media:** Contains precursors which means that the macromolecules need to be built (anabolism)
- **Complex media components:** Contains macromolecules so the target product is produced faster.

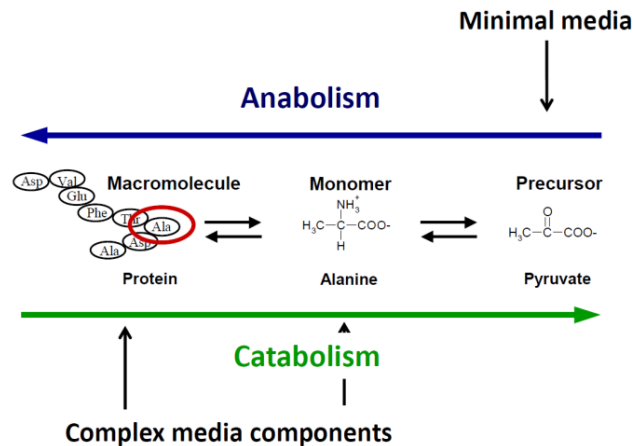


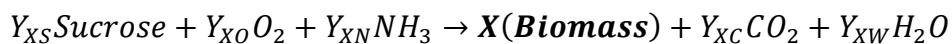
Figure 1: Simplified representation of the metabolic pathways involved in the synthesis and degradation of biomolecules.

To optimize the process, we need to use pure materials.

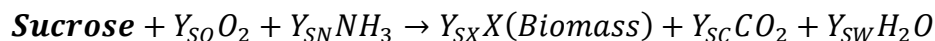
3.5 Principal components of fermentation media

We can establish an equation with stoichiometry

- Reaction equation using the biomass as reference



- Reaction equation using the substrate as a reference



We look at the cell composition to formulate the media, using this equation we can calculate the minimal amount of nutrients to produce the required product yields.

Some components are not essential but might be useful depending on the case.

- Chelating agents
- Buffers
- Inhibitors/inducers

When choosing the components it's important to have in consideration the foaming effect.

Foam is spooky

3.5.1 Water

Water is used in multiple contexts in fermentation: Growth medium, temperature control, steam sterilization and cleaning. Water needs to have the right quality or pre-treatment is required.

Water consumption should be limited since water is expensive and the pre-treatment too.

3.5.2 Carbon sources

When designing a media, the choice of carbon source is influenced by several factors:

- Metabolite (glucose) repression: High glucose levels can inhibit the use of other carbon sources.
- Product formation: The carbon source should support the desired product yield.
- Economy of the process: The cost-effectiveness of the carbon source.
- Supply of carbon source: Availability and consistency of the carbon source.
- Purity: The carbon source should be free from contaminants.
- Method of sterilization: The chosen method should not degrade the carbon source, glucose caramelizes when sterilized by heat.
- Metabolism: Glucose might change the metabolism of the microorganism so this effect need to be know previously.

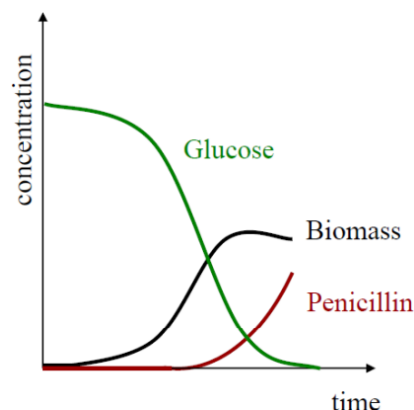


Figure 2: Typical phases of a fermentation process, including the lag phase, exponential growth phase, and stationary phase, where secondary metabolites like penicillin are produced.

Different carbon sources are available:

- **Sucrose**
 - Source: Beet and sugar cane
 - Purified or used as molasses (e.g insulin production)
- **Lactose**
 - Source: Whey (by-product of dairy operations)
 - No catabolite repression
- **Starch**
 - Source: corn, potatoes...
 - Easily broken down by filamentous fungi
 - Used as it is or pretreated to release sugar monomer

- **Corn steep liquor**
 - Nitrogen source also containing carbons
- **Cellulose**
 - Source: wood, forest and agricultural waste
 - Hard to pretreat in order to release sugars
 - Spread use for bioethanol production.

Some other alternative Carbon sources where the usage depends on the microorganism:

- **Oils and fats**
 - Oil contains x2.4 energy compared to glucose
 - Has anti-foam properties
 - Some oil might be essential in case it can't be synthesized by the cell.
 - However, oxygen might not be that well distributed.
- **Hydrocarbons**
 - Methanol, methane...
 - Price is unstable
 - Releases high amount of heat so more cooling is required.

The more reduced is a carbon source, the higher the need of oxygen.

3.5.3 Nitrogen Sources

- **Inorganic**
 - Ammonia gas is the most used one but since it's in gas form, it helps with mixing but the bubbles generate foam (and it's evil)
 - Ammonium salts
 - Nitrates
- **Organic (complex media)**
 - Amino acids, proteins, complex types, urea (influences metabolism in filamentous fungi.
 - Soybean hydrolysate
 - Corn-steep liquor
 - Peanut powder
- **Factors that influence choice**
 - pH control
 - Control of metabolism
 - Possible problems with purification

3.5.4 Minerals

Minerals need to be dissolved. Magnesium is needed for DNA production and can precipitate while sterilization.

- Micronutrients, often found as impurities in other media components /chemicals
- Macronutrients are to be added (Mg, P, K and S)

3.5.5 Oxygen

Oxygen must be dissolved so the cell can use it, getting enough oxygen dissolved is a major problem in industrial fermentations. Oxygen determines:

- Cell growth
- Rheology: Viscosity and flow properties of the culture broth.
- Antifoam chemicals: Increases surface tension and lowers O₂ distribution.

3.5.6 Non-essential in media

- Chelating (e.g EDTA)
 - Might be necessary to prevent precipitation (mainly for minerals)
 - Not needed for all minerals
- Growth promoting chemicals
 - Especially interesting when using minimal media.
 - Chemicals that cells can't synthesize
 - Macromolecules to make growth faster.
 - Many can be found in complex C and N sources
 - May be necessary to add to get growth, especially in bacteria.
- Buffers
 - To control the pH of fermentation (phosphate salts in growing media)
- Precursors
 - The precursors for final product are added in order to achieve the highest production rate.



- Inhibitors
 - Inhibit specific metabolism pathways to direct anabolism. When inhibiting a specific enzyme, another pathway is “unlocked” producing another product.
 - An inhibitor might not inhibit a specific pathway.
- Inducers
 - To induce the production of a specific enzyme or protein
 - Can be used for separating growth from product formation
 - Can be expensive
 - Important for pharma and industrial enzymes.

3.6 Foam

Foam is always produced due to CO₂ production but can also appear due to extra O₂ supply. It is also formed when proteins are released in the fermenter, problematic if cells die.

- Can reduce the volume of the liquid that can be used in the fermenter.

- It gives problems when maintaining sterile environment since it can go through feeding supply.
- Most fermentations need antifoam agents.
- Some cells get trapped in foam which stress them and makes them bust. Busting is releasing proteases that generate even more foam.

Antifoams are chemicals which don't interfere with the cells and increase the surface tension, these typically consists of:

- Alcohol, stearyl and octyl decyl
- Esters
- Fatty acids
- Silicone

Lecture 4: Mass balances and kinetics (Batch)

Batch fermentation is a process where all nutrients are added to the bioreactor at the beginning of the fermentation. The system remains closed, and growth is not limited initially, allowing the cells to grow at maximum rate ($\mu = \mu_{max}$) and the process enters the stationary phase. This process is widely used in industries for producing antibiotics, enzymes and other bioproducts due to its simplicity and effectiveness.

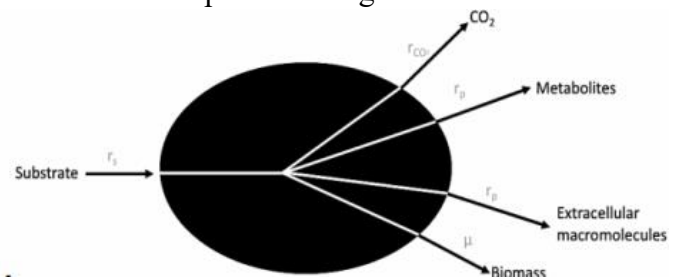
4.1 Quantitative description of microbial physiology

- Yield: How efficiently cells convert substrate into biomass or products.
- Productivity: The rate at which products are formed.

4.1.1 Black box model

This is the simplest model possible that contains all main components for growth:

- Carbon (cells walls, biomass)
- Hydrogen (protons)
- Nitrogen (aa, biomass)
- Phosphate (DNA/RNA)



$$Element_{in} = Element_{out}$$

4.2 Stoichiometric Equation

The stoichiometric equation is a more detailed representation of the fermentation process. It accounts for the main components involved in the reaction: the substrate (S), biomass (X), and products (P). The general form of the stoichiometric equation is:



$Y_S S, Y_N N, Y_O O, Y_{SN} S_S, Y_X X, Y_C P CO_2, Y_P P$ = stoichiometric coefficient

4.3 Fermentation modes

There are three main modes of fermentation: batch, continuous, and fed-batch. Each mode has its advantages and disadvantages, and the choice of mode depends on the type of product being produced.

- **Batch Fermentation:** In this mode, all nutrients are added at the beginning, and the fermentation runs until the substrate is depleted. It is versatile and has a low risk of contamination, but it requires significant downtime for cleaning and sterilization between batches.

- **Continuous Fermentation:** Nutrients are continuously added, and products are continuously removed. This mode offers high efficiency and constant product quality but is prone to contamination and requires careful control of the process.
- **Fed-Batch Fermentation:** Nutrients are added incrementally during the fermentation, allowing for high cell densities and controlled conditions. It combines some advantages of both batch and continuous fermentation but still faces challenges related to contamination and process control.

4.4 Batch fermentation

4.4.1 Specific growth rate μ

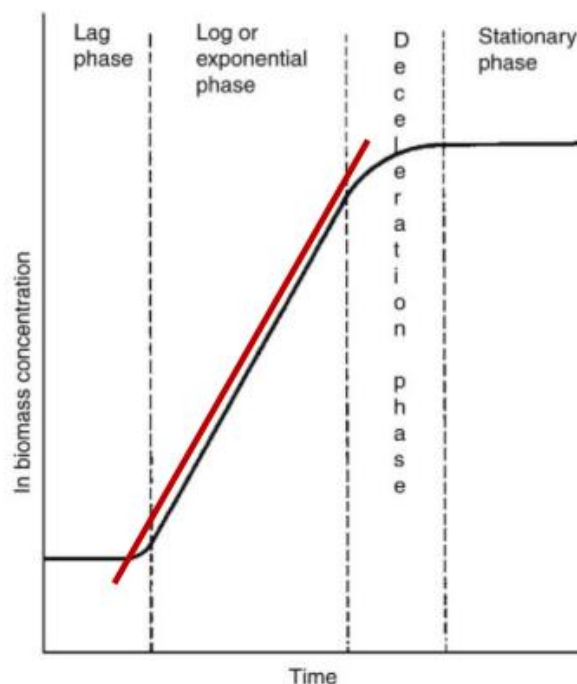
The specific growth rate (μ) is a key parameter in batch fermentation. During the log phase, cells grow exponentially, and the growth rate can be described by the equation:

$$\frac{dx}{dt} = \mu x$$

Integrating this equation gives:

$$x_t = x_0 e^{\mu t}$$

By plotting the natural logarithm of biomass concentration ($\ln x$) against time, the slope of the line represents the specific growth rate (μ).



4.4.2 Yields

Yield coefficients describe how efficiently the substrate is converted into biomass or products. **Yields are always given with a double index which indicates the direction of the conversion**

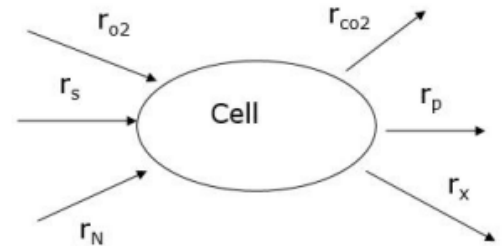
For example,

- Y_{sx} represents the amount of **substrate used per unit of biomass** formed.
- Y_{xp} represents the amount of **product formed per unit of biomass**.

4.4.3 Productivity

Productivity is a measure of how quickly products are formed. Volumetric productivity (q) is expressed in grams per liter per hour (g/L/h), while specific productivity (r) is expressed in grams per gram of dry weight per hour (g/g DW/h).

- r_s specific substrate consumption rate
- $r_x = \mu$ specific growth rate
- r_p specific rate of product formation (r_e, r_a, r_{gly})
- r_{CO_2} is the specific CO_2 production rate (CER)
- r_{O_2} specific oxygen consumption rate (OUR)

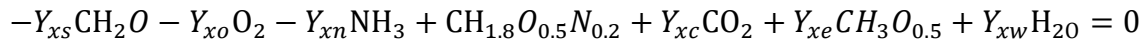


4.5 Linking yields and stoichiometry

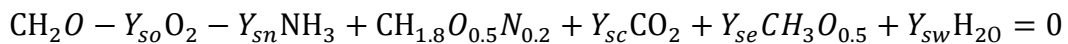
The stoichiometric equation can be used to link yields and understand how substrates are converted into biomass and products.

For example, in respiratory-fermentative growth with *Saccharomyces cerevisiae*, the stoichiometric equation can be normalized to biomass or substrate to calculate missing yields.

- Normalized to biomass



- Normalized to substrate



Example:

$$\frac{Y_{xe}(\text{Cmol EtOH} \cdot \text{Cmol DW}^{-1})}{Y_{xs}(\text{Cmol Glc} \cdot \text{Cmol DW}^{-1})} = Y_{se}(\text{Cmol EtOH} \cdot \text{Cmol Glc}^{-1})$$

4.6 Monod kinetics

Monod kinetics describe how the specific growth rate (μ) depends on the substrate concentration (C_s)

where μ_{max} is the maximum growth rate, and K_s is the half-saturation constant. At low substrate concentrations, the growth rate is proportional to C_s , while at high concentrations, the growth rate approaches μ_{max} .

- Substrate inhibition: Occurs when high concentrations of the substrate (e.g., glucose) **slow down** microbial growth.

$$\mu = \mu_{\max} \frac{c_s}{c_s^2/K_i + c_s + K_s}$$

Here, K_i is the inhibition constant.

- At low substrate concentrations ($C_s \ll K_s$), the growth rate increases with c_s
- At high substrate concentrations ($C_s \gg K_s$), the term C_s^2/K_i becomes dominant, reducing the growth rate.

- Product inhibition: Occurs when the accumulation of metabolic products (e.g ethanol) in the bioreactor **inhibits microbial growth**. These can be toxic or interfere metabolic pathways.

$$\mu = \mu_{\max} \frac{c_s}{c_s + K_s} \frac{1}{1 + c_p/K_i}$$

Here, c_p is the product concentration.

- Here the first term describes the growth rate as a function of substrate concentration (like Monod kinetics)
- The second term reduces the growth rate as the product concentration increases

4.6.1 Maintenance coefficient

Maintenance refers to the energy and resources necessary for a cell to stay alive and maintain basic cellular functions, such as:

- Gradients and electrical potential: Maintaining ion gradients across cell membranes.
- Futile cycles: Metabolic pathways that consume energy without production useful work.
- Turnover of macromolecules: Degradation and resynthesis of proteins, RNA, and other cellular components.

This is important because even when cells are not growing, they still consume substrate to maintain cellular functions. Thus, the Y_{sx} is not constant because some of the substrate is used for maintenance.

- **Pirt-Maintenance Equation:**

$$r_s = \frac{Y_{true}}{Y_{xs}} \mu + m_s$$

Where Y_{true} is the true Yield coefficient, representing the max theoretical yield of biomass if there were no maintenance requirements. m_s is the maintenance coefficient, representing the rate of substrate consumption for maintenance.

Knowing that,

$$r_s = \frac{Y_{true}}{Y_{xs}} \mu + m_s \quad r_s = Y_{xs}$$

We can find the observed yield by:

$$Y_{sx} = \frac{\mu}{Y_{xs}^{true} \mu + m_s}$$

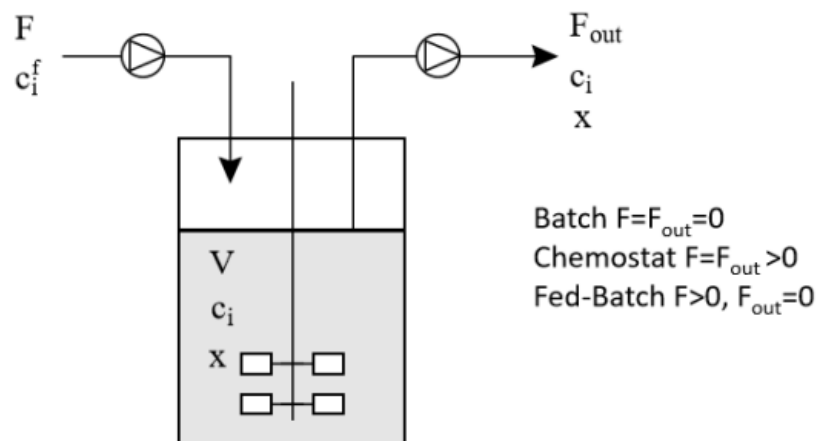
- Y_{sx} represents the amount of biomass produced considering both growth and maintenance.
- Y_{xs}^{true} The true yield coefficient is the max theoretical yield if there were no maintenance requirements.
- m_s is the maintenance coefficient.

We can observe how the numerator represents the amount of biomass being produced and the denominator is the total substrate consumption, divided into two parts: Growth and maintenance.

Key observations,

- Y_{sx} is not constant
- Y_{sx} decreases at low specific growth rates. A larger fraction of the substrate is used for maintenance.
- Y_{sx} approaches the reciprocal of Y_{xs}^{true} for high μ , since the impact is minimized and the observed yield approaches the theoretical maximum yield.

4.7 Mass balances for bioreactors



$$\text{Accumulated} = \text{In} - \text{Out} + \text{Produced} - \text{Consumed}$$

4.7.1 General balances

The mass balances for substrate, product and biomass are:

- Substrate: The rate of change of substrate concentration over time, it changes due to the consumption and flow.

$$\frac{dc_{s,i}}{dt} = -r_{s,i}x + D(c_{s,i}^f - c_{s,i})$$

- Product: The rate of change of product concentration over time, it changes due to production and flow.

$$\frac{dc_{p,i}}{dt} = -r_{p,i}x + D(c_{p,i}^f - c_{p,i})$$

- Biomass: The rate of change of biomass concentration over time, it changes due to the growth.

$$\frac{d(x)}{dt} = \mu * X$$

- ❖ $c_{s,i}^f$ is the substrate/ product concentration in the feed and $c_{s,i}$ is the substrate concentration in the bioreactor.
- ❖ $r_{s,i}$ is the specific substrate consumption and x is the biomass concentration.
- ❖ D is the dilution rate ($D=F/V$) where F is the flow rate and V is the volume of the bioreactor.

$$D = \frac{F}{V}$$

4.7.2 General balances for Batch

However, all these formulas are for general balances, for **batch specifically** there's no inflow or outflow so $D=0$. Thus,

- Substrate:

$$\frac{dc_{s,i}}{dt} = -r_{s,i}x$$

- Product:

$$\frac{dc_{p,i}}{dt} = -r_{p,i}x$$

- Biomass:

$$\frac{d(x)}{dt} = \mu X$$

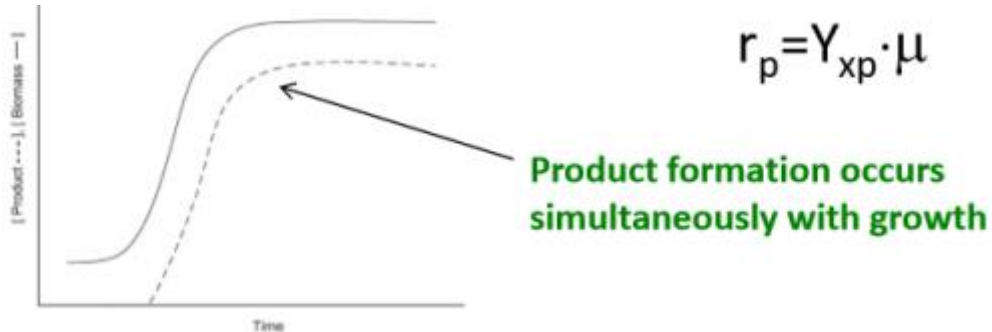
4.8 Calculating productivities

4.8.1 Productivity during growth

In many fermentation processes, the product is formed **simultaneously** with microbial growth. The productivity can be estimated by analyzing the relationship between the product concentration (p) and the biomass concentration (x).

$$r_p = Y_{xp} * \mu$$

- r_p : Specific productivity (g product/g biomass/h).
- Y_{xp} : Yield of product on biomass (g product/g biomass).
- μ : Specific growth rate (h^{-1}).



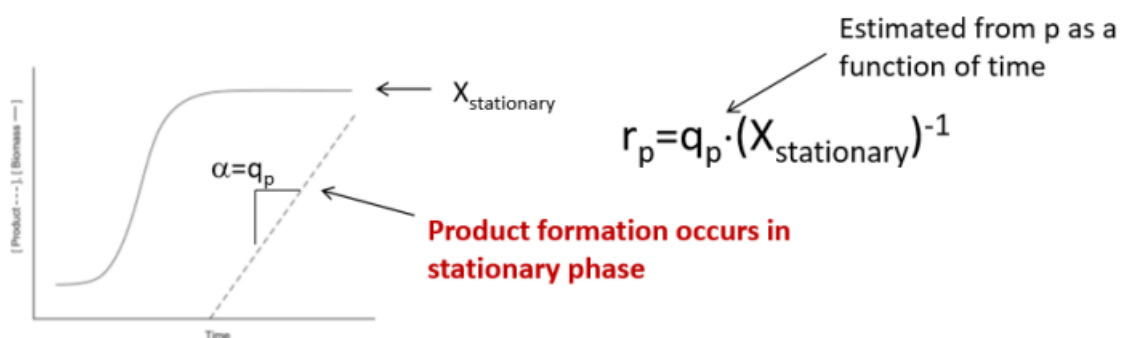
4.8.2 Productivity during the stationary phase

In some fermentation processes, the product is formed primarily during the **stationary phase**, when the cells are no longer growing but are still metabolically active. This is common in processes where the product is a secondary metabolite (e.g., antibiotics, ethanol).

Here we can analyze the product concentration (p) as a function of time.

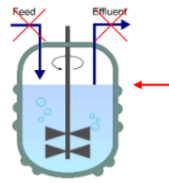
$$r_p = \frac{q_p}{X_{stationary}}$$

- r_p : Specific productivity (g product/g biomass/h).
- q_p : Volumetric productivity (g product/L/h).
- $X_{stationary}$: Biomass concentration during the stationary phase (g biomass/L).

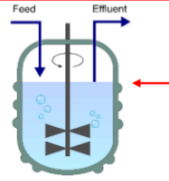


Lecture 5: Mass Balances and Kinetics (Chemostat)

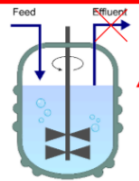
- Batch



- Continuous



- Fed-batch



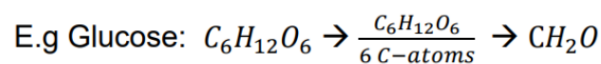
Chemostat is not exactly the same as continuous fermentation; rather, a chemostat is a type of continuous fermentation where one of the nutrients (usually carbon) is the limiting factor, controlling the growth rate.

5.1 Carbon moles (C-moles)

- When using the carbon-mole concept, we focus on the number of carbon atoms. This helps calculate how much substrate (like glucose) goes to every product.
- **Example:** Glucose and galactose have same carbon atom numbers, despite having the different weight. By using carbon-moles, we can account for this difference in fermentation equations.

5.1.1 Stoichiometry:

All products are based on the amount of carbon. We calculate how many carbons per mole to ensure balance. For example, in glucose, there is 1 carbon unit for every 2 hydrogen and oxygen atoms.



$$\text{Ethanol: } C_2H_6O \text{ divided by 2 C-atoms } CH_3O_{0.5} \quad CM_W = \frac{46.8 \frac{g EtOH}{Mole EtOH}}{2 \text{ C-atoms}} = 23.4 \frac{g EtOH}{Cmole etOH}$$

$$Y_{SEtOH} = 0.1 \frac{Cmole EtOH}{Cmole glc} * \left(\frac{23.4 \frac{g EtOH}{Cmole etOH}}{30 \frac{g glc}{Cmole glc}} \right) = 0.077 \frac{g EtOH}{g glc}$$

$$Y_{SEtOH} = 0.077 \frac{g EtOH}{g glc} * \left(\frac{30 \frac{g glc}{Cmole glc}}{23.4 \frac{g EtOH}{Cmole etOH}} \right) = 0.1 \frac{Cmole EtOH}{Cmole glc}$$

5.2 Continuous Batch (Chemostat)

In a continuous fermentation, media is added continuously, and a mixture of biomass, media, and products is removed at a constant rate.

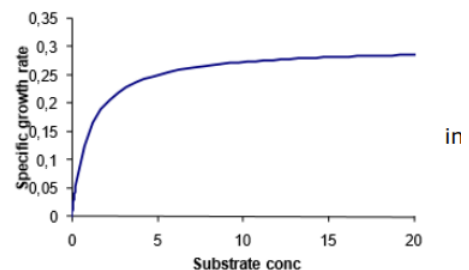
- **Flow rates:** The rate of media coming in and going out must remain equal for the system to be in equilibrium.

More specifically, **chemostat fermentation** is when one of the components (typically carbon) is limiting, so we can control the growth rate. Concentrations of the substances need to remain constant. The period before these concentrations stabilize is not considered part of the chemostat process. There are very few industries that use the chemostat process since their main goal is the product concentration but the insulin production at Novo Nordisk is achieved by chemostat.

5.2.1 Growth Dynamics

At low C_s (substrate concentration), the growth rate is proportional to C_s . When the substrate concentration exceeds a certain level, the maximum growth rate (μ_{max}) is achieved, and further increasing the substrate concentration does not result in faster growth.

$$\mu = \mu_{max} \frac{C_s}{C_s + K_s}$$



- μ : Specific growth rate of the microorganisms (units: h^{-1} or s^{-1}).
- μ_{max} : Maximum specific growth rate achieved when the substrate concentration is very high.
- C_s : Substrate concentration in the bioreactor.
- K_s : Half-saturation constant, which represents the substrate concentration at which the growth rate is half of μ_{max} . It is a measure of the affinity of microorganisms for the substrate.

5.2.2 Chemostat Mass Balances

- In a chemostat, the change in mass over time can be described by:

$$\frac{dx}{dt} = \mu x - Dx$$

where:

- μ is the specific growth rate
 - D is the dilution rate
 - x is the biomass concentration
- Under steady-state conditions, there is no change in mass over time:

$$\frac{dx}{dt} = 0 \quad \text{thus} \quad D = \mu$$

- This means that the **dilution rate** equals the growth rate.

5.2.3 Dilution Rate and Critical Dilution Rate

- The dilution rate is controlled by the operator and is calculated as:

$$D = \frac{F}{V}$$

where F is the flow rate (L/h), and V is the vessel volume (L).

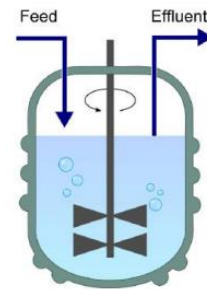
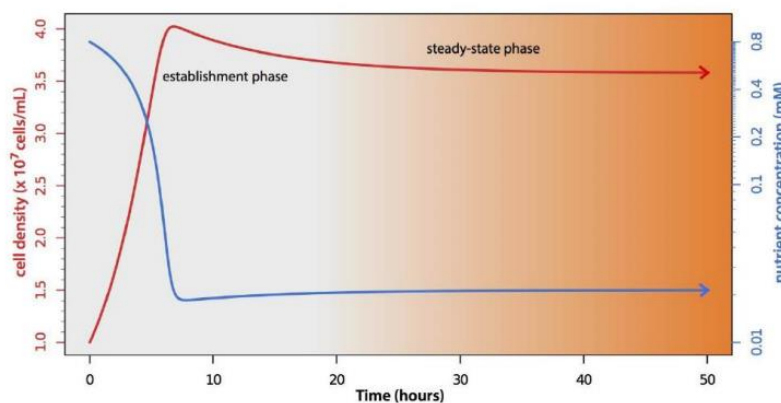
- The **critical dilution rate (D_{crit})** is the maximum dilution rate where the cells can still grow. If the dilution rate exceeds this critical value, the cells are washed out faster than they can grow, leading to cell washout.

5.2.4 Residence Time

Residence time (or retention time) is the time it takes for all the volume inside the bioreactor to be replaced, calculated as:

$$RT = \frac{1}{D}$$

It typically takes about 5 residence times for the system to reach a steady state. If the system hasn't stabilized after this, something might be wrong with the design.



5.3 Steady State vs. Stationary Phase

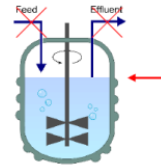
- **Steady state** (chemostat): There is continuous exponential growth, where we determine the speed, with a constant cell concentration because the substrate is replenished.
- **Stationary phase** (batch fermentation): There is no growth as the substrate has been fully consumed, and this phase only occurs in batch processes.

5.4 Industrial Application and Research

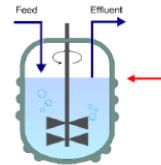
- Chemostats are not commonly used in industry, with some exceptions like insulin production at **Novo Nordisk**.
- They are more frequently used in research, for instance, to calculate constants such as affinity constants for substrates.
- Chemostats are also used to study microbial maintenance, which refers to the energy required for the cell to stay alive without growing.

Lecture 6: Mass balances and kinetics (fed-batch)

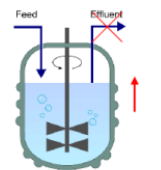
- Batch



- Chemostat



- Fed-batch



In fed-batch fermentation, the process begins with a full tank. At a certain point, fresh media is added to the tank, but no material is removed. This differs from a chemostat, where material is continuously added and removed. Fed-batch fermentation is a finite process because the tank eventually reaches its maximum capacity.

- **Volume and Mass Balances:** Over time, the volume in the tank increases due to the addition of fresh media. This complicates calculations, as both the added volume and the increasing tank volume must be accounted for. Mass balances must be calculated over time using differential equations, as they change continuously.

6.1 Fed-Batch

6.1.1 Key variables in Fed-batch fermentation

- Volume: $V = V(t)$
- Cell concentration: $x = x(t)$
- Product concentration: $c_p = c_p(t)$

6.1.2 Fed-Batch qualities

The main goal of fed-batch fermentation is to achieve the **highest possible concentration of the desired product**.

Like a chemostat, the **growth rate of cells can be controlled** by regulating the amount of carbon source added to the tank. This helps avoid the inhibitory effects of high substrate concentrations.

- **Substrate inhibition:** High concentrations of substrates like glucose can inhibit cell growth and lead to the production of unwanted by-products, such as ethanol in yeast fermentation. Fed-batch fermentation allows for the gradual addition of substrate, maintaining an optimal concentration and preventing inhibition.

In a batch process, the amount of carbon source is limited to what is added initially. In contrast, fed-batch fermentation allows for **continuous addition of carbon**, enabling higher product yields.

6.2 Industry applications

While many industrial processes use batch fermentation, fed-batch is the most commonly used method for producing high-value products.

Fed-batch fermentation aims for **high respiration rates** to prevent the production of secondary, non-desired by-products like ethanol. It also targets a **high yield of cells** (Y_{sx}), as higher cell concentrations can lead to increased product formation, though this may also induce secondary metabolite production.

As the volume changes, the concentrations also change, if we control the flow (F) in order to keep them constant, we can reach a quasi-steady-state (in this class it can be assumed that it is a steady-state)

6.2.1 Example for Novo Nordisk

The next process describes how insulin is produced at Novo Nordisk, using the three different types of fermentations that we've studied.

Novo starts with a fresh culture in a wide-bottom bottle containing a hard medium. The culture is grown to obtain pure cultures, which are then used to inoculate a batch fermentation. Once the glucose in the batch is depleted, the process transitions to fed-batch fermentation to ensure a high concentration of the seed culture. This fed-batch culture is then used to seed a chemostat tank.

6.3 Challenges in Fed-Batch fermentation

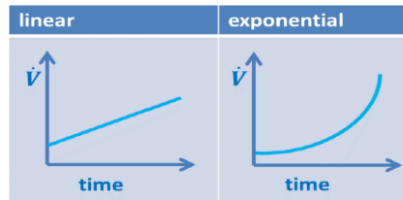
6.3.1 Decreasing glucose concentration

As fresh media is added, the volume in the tank increases, causing the glucose concentration to decrease over time. The dilution rate (D) is given by:

$$D = \frac{F}{V}$$

To maintain a constant dilution rate and steady cell growth, the flow rate (FF) can be increased. However, this also increases the volume, creating a loop that requires continuous adjustment. This is known as **exponential feeding**.

6.4 Feeding profiles



6.4.1 Exponential feeding

The amount of fresh media is increased over time.

- Control of Growth Rate: Exponential feeding allows for precise control of the culture's growth rate, maintaining a quasi-steady state.
- Flow Rate Calculation: The flow rate for exponential feeding can be calculated using the following formula:

$$F(t) = F_0 * E^{\mu t}$$
$$F_0 = \frac{\mu * x_0 * V_0}{Y_{sx}(C_s^f - C_s)}$$

- F_0 is the initial flow rate,
- μ is the specific growth rate,
- t is time.
- x_0 is the initial cell concentration,
- V_0 is the initial volume,
- Y_{sx} is the yield coefficient,
- C_{sf} is the substrate concentration in the feed,
- C_s is the substrate concentration in the tank.

6.4.2 Linear feeding

The amount of fresh media added is constant.

- Secondary metabolites: Linear feeding is used when we want to produce secondary metabolites. These metabolites are usually produced when the carbon is limited, and the cells grow slower.
- Oxygen limitation: Linear feeding can be used to make sure we don't get overgrown that can lead to oxygen limitation, terminating the fermentation earlier.

Reactor	Advantages	Disadvantages
Batch	Versatile, since it can be used for many different processes Low risk of contamination. Complete conversion of substrate possible	High labor cost Much idle time, due to cleaning and sterilization after each fermentation
Continuous	High efficiency of the reactor capacity High productivity can be maintained for long period of times Automation is simple Constant product quality	Problems with infections Possibility for appearance of low producing mutants during long operations Inflexible since it can rarely be used for different processes without substantial retrofitting Downstream processing has to be adjusted to the flow through the bioreactor (or holding tanks are required)
Fed-batch	Allows operating at well controlled conditions by controlling the feed addition Allows very high cell densities and hereby high final titers	Some of the same problems as for the batch and continuous reactor, but generally the disadvantages are less pronounced with this mode of operation

Lecture 8: Introduction to bioreactors

8.1 Components of a bioreactor

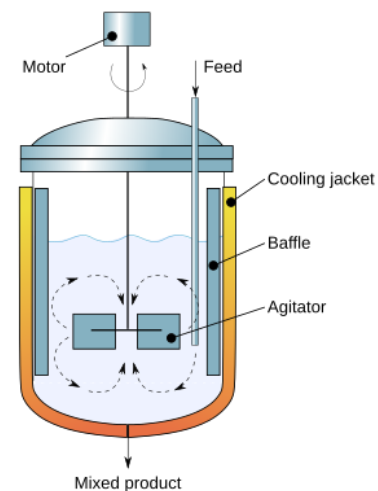
8.1.1 The vessel and thermal jacket

- The **vessel** is the container that holds the microbial culture.
- The **thermal jacket** keeps the correct temperature, it can both heat or cool down the vessel. It can be a vest that is applied around the tank or it can be integrated in the tank.

Microbial processes generate heat, that's why we need to regulate the temperature.

8.1.2 Agitation system

- **Impellers:** Rotating blades or paddles inside the bioreactor. These make sure the elements are mixed, enhances mass transfer and prevent sedimentation.
 - Pitch-blade, for fragile cells.
 - Rushton, for microbial with high oxygen need.
- **Baffles:** Vertical strips fixed in the inner side of the vessel walls. These prevent vortex formation caused by the impeller and promote turbulent flow to improve mixing efficiency.
- **Motor:** Just to power the impeller



8.1.3 Gas supply

The type of gas used depends on the microorganism and the process requirements. It is delivered using a filter that ensure sterility connected to a sparger that creates small bubbles. The bubbles have to be small in order to increase the surface area/volume ratio, facilitating gas transfer.

Fun fact: sparger is also used to displaced gases in liquid media. For example: if we want a media to be anaerobic, we can use an sparger with N_2 to displace the oxygen out of the media.

8.1.4 Condenser

We will have volatile compounds so we use the condenser to prevent evaporation by condensing the gasses so they dilute again in the mixture.

8.1.5 Probes

These can measure many variables and the readings are used to monitor and control. The most common ones are:

- Temperature
- pH
- DO (dissolved oxygen): These don't measure absolute O_2 concentration, only relative O_2 (partial pressure)
 - Polarographic sensor
 - Optical sensor

Feature	Polarographic Sensor	Optical Sensor
Principle	Electrochemical	Fluorescence quenching
Oxygen consumption	Yes	No
Warm-up time (polarization)	Required	None
Maintenance	Higher	Lower
Calibration frequency	Frequent	Infrequent
Performance at low DO	Less accurate	High accuracy
Ideal for	Traditional, cost-sensitive	Modern, low-maintenance

The DO measurement in ST (standard benchtop) bioreactors are affected by:

- Temperature, pH, agitation, aeration, sterilization.

8.1.6 Sampling

We want to sample in order to do off-line measurements and keep track of the evolution of the culture. The most common set up is a tube connected to syringe that extracts culture volume. After that, gas need to be added through the top syringe in the image below to ensure pressure balance.



8.1.7 Headplate

It's the lid that closes the system, it has holes to accommodate all the fittings:

- Probes
- Sampling tube
- Inoculation port
- 4-way port (C source...)

8.1.8 The tower and the control panel

- Tower: Gas supply, cooling water supply, process connection and all the pumps.
- Control panel: Display and control of parameters and probes calibration.

8.2 Bioreactors in the laboratory

A bioreactor is roughly anything with microorganisms growing inside of it. Official definition: A system that keeps an active biological environment.

There's different systems with different scales:

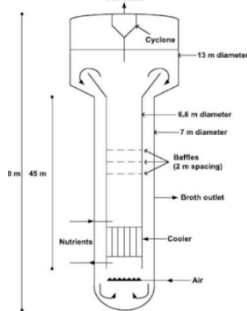
- Uncontrolled: Shake flasks, microtiter plates
- Controlled: Ambr systems, micro fermentation systems:
 - Growth profiler: Can run up to 960 fermentations which makes them good for screening, can regulate temperature and shaking and reduce evaporation.
 - BioLector: Up to 48 fermentations, more variables and control like temperature, shaking, humidity, pH... Can measure biomass, fluorescence, pH... It can also control the atmosphere composition so it's good for **anerobic cultivation**.

Feature	Growth Profiler	Traditional Fermentation Systems (e.g., Shake Flasks)
Measurement Method	Backscatter or fluorescence	Absorbance (OD600)
Sample Volume	Small (100-300 µL per well)	Larger (e.g., 50-500 mL)
Throughput	High (up to 96 cultures at once)	Low (1 culture per flask)
Contamination Risk	Low (lid remains closed)	Higher (requires sampling for OD readings)
Evaporation Control	Minimal evaporation	Higher evaporation, especially during long experiments

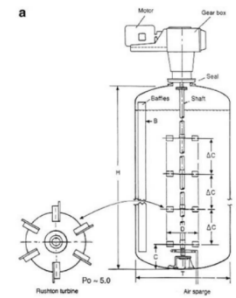
Feature	BioLector	Traditional Systems (e.g., Shake Flasks)
Throughput	High (48-96 parallel cultures)	Low (1 flask per culture)
Non-Invasive Monitoring	Yes (real-time measurements)	No (requires sampling)
Measured Parameters	OD, pH, DO, fluorescence	OD only (usually)
Culture Volume	Small (200-1000 µL)	Large (50-500 mL)
Aeration Control	Enhanced (shaking and gas exchange)	Limited (depends on flask design)
Automation	Available in advanced models (BioLector Pro, RoboLector)	Rare (requires manual operation)

8.3 Bioreactors outside of the laboratory

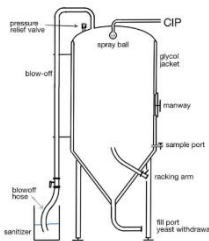
- **Stirred tanks:** The most simple set up, large amount of volume.



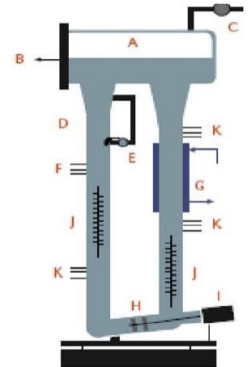
- **Air lift bioreactors:** Tall and slim, without mechanical stirring since bubbles are used to mix. Gas is sparged at the bottom making the fluid circulate up.



- **Forced flow loop (U-loop):** Recirculation by pumps (high power input), very efficient gas transfer, easy temperature control, excellent for high demanding aerobic cultivations.



- **Cylindroconical fermenters:** Brewing industry, work under pressure, easy yeast removal, thermal jacket, no mechanical mixing.



- **Photobioreactor:** Converts sunlight into chemical energy, used to remove nutrients and suspended solids, produces valuable compounds like pigments, proteins...

Feature	Packed bed bioreactor	Fluidized bed bioreactor
Structure	Filled with solid packing material (e.g fibers)	Small immobilized particles suspended by fluid flow
Mixing	Minimal or no mixing (uses liquid flow)	Excellent mixing due to fluidized motion
Applications	- Wastewater treatment - Immobilized enzymes/cells - Efficient substrate use	- Wastewater treatment - Fermentation - Bioremediation
Advantages	- High cell density - Low shear stress - Efficient substrate use	- Higher efficiency - Reduced clogging - Larger surface area.
Limitations	- Risk of clogging - Uneven nutrient distribution	- Required precise flow control to avoid particle washout.

Lecture 9: Microbial cell factories and strain improvement

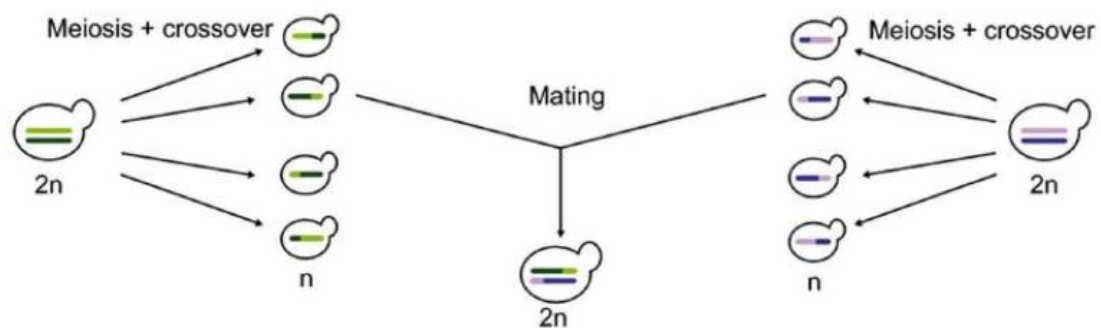
When talking about strain improvement, we refer to a set of features. These can be:

- Formation of new products
- Product yield, titer and productivity
- Reduction of by-product formation
- Growth rate
- Substrate utilization/tolerance
- Product tolerance
- Strain robustness
- Strain stability

9.1 Classic strain improvement

9.1.1 Origin of natural diversity

In classical strain improvement, the original strain naturally generates genetic diversity that creates new isolates that through screening or selection are isolated.



Genetic recombination

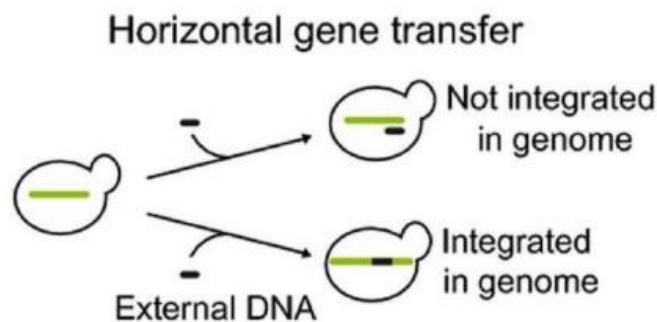


In nature, the origin of natural diversity is due to spontaneous mutations. These occur by errors in the DNA duplication, either point mutations, insertion / deletion, transposons...

Some microorganisms might undergo variations in ploidy level, contributing in diversity, a typical example is *Saccharomyces cerevisiae*.

- Types of ploidy changes:
 - **Whole Genome Duplication (WGD):** The entire genome is duplicated (e.g., diploid → tetraploid). *S. cerevisiae* underwent WGD ~100 million years ago, leading to evolutionary innovations.
 - **Chromosomal Duplication:** Single chromosomes are duplicated (e.g., Chr III in yeast). Can provide extra copies of genes for adaptation.
 - **Aneuploidy:** Gain/loss of individual chromosomes (e.g., diploid +1 chromosome). Common in industrial yeast strains under stress (e.g., high temperature, ethanol).
 - **Gene Duplication:** Specific genes are copied (e.g., ADH1 and ADH2 in yeast). Allows functional divergence (e.g., ADH1 converts acetaldehyde → ethanol; ADH2 does the reverse).

In some cases, external DNA can be integrated in the genome of a microorganism (integrated in the chromosome) or not integrated (DNA exists temporary e.g plasmids). This is horizontal gene transfer.



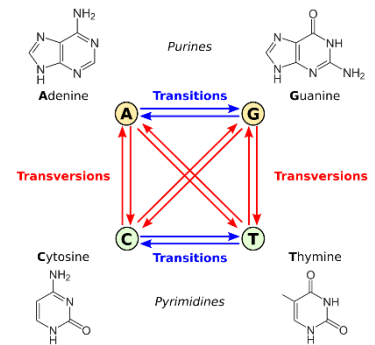
There is really low genetic diversity among the industrially used strains while the potential in wild strains is enormous.

9.1.2 Artificial diversity

Referred as random mutagenesis, this uses either chemical or physical mutagens:

- Chemical mutagens:
 - Base analogues: Mimic natural DNA bases to make subtle changes that may alter protein function.

- Removal or addition of functional groups: Causing transitions or transversions. This generates more drastic mutations.
- Physical mutagens:
 - UV: Error-prone repair introduces mutations.
 - Ionizing radiation: Breaks DNA strands, leading to large deletions, translocations, or chromosome loss.



9.1.3 Screening

The easiest situation is a yes/no answer. This is growth or no growth, pigment production...

Then we can get some that require quantitative analysis:

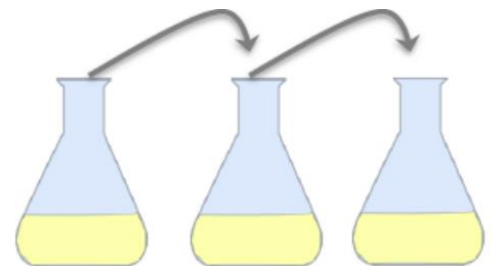
- Growth (tolerance to compounds)
- Flocculation
- Production of specific compound

We can select by: utilization of specific substrates, co-consumption of substrates, substrate tolerance, stress tolerance, product tolerance...

9.1.4 Selection

We can do it by a serial transfer of the strain over many generations. Here is the environment that selects the strains.

Here we would take a strain with optimal mutagenesis and put it in the first flask with some evolutionary pressure (pH, substrate, toxic metabolite...)



- Now we collect the alive cells and put them in the next flask and repeat until we get a better strain.

We can also use a chemostat to select since it maintains a constant environment where only the best-adapted microorganisms can survive.

Depending on what we are selecting for we will use the flask method (lag phase and growth rate) or the chemostat method (growth rate and substrate affinity)

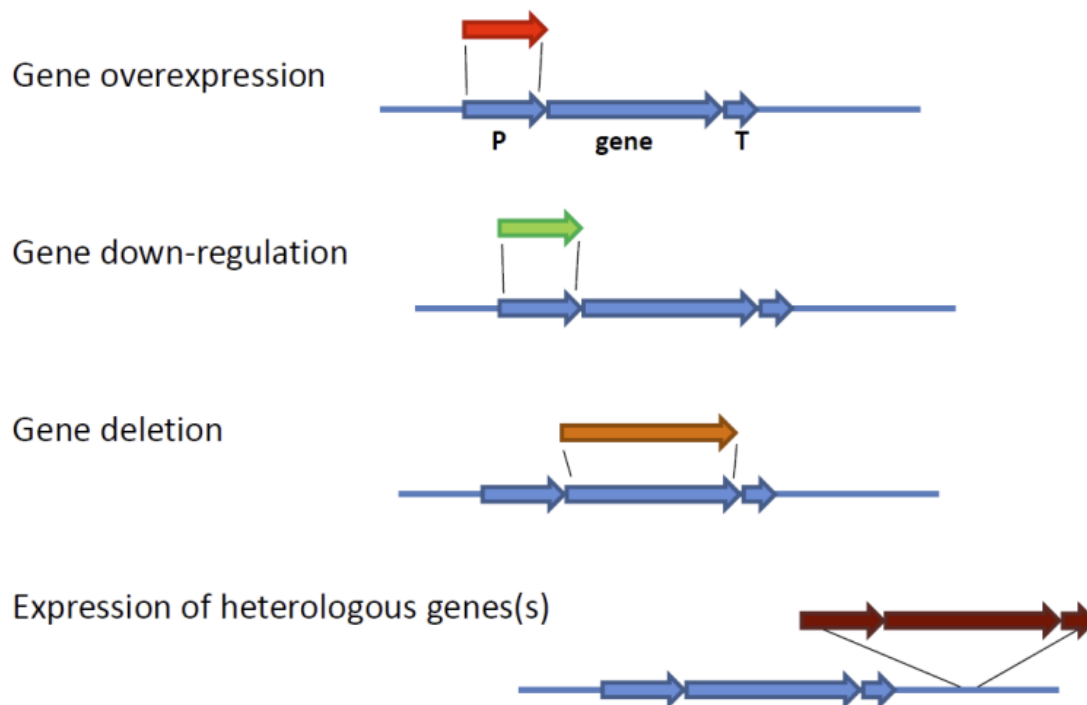
Advantages	Disadvantages
Use of huge potential of natural diversity	Traits limited to natural diversity
Lower knowledge needs on genetic basis - Suitable for complex traits - Less-characterized strains	Screening for certain traits is difficult Many industrial strains not suitable for crossing
Mostly classified as non-GMO	Acquisition of trade-offs

With regards to the lower knowledge points, we don't really need to know why a strain is better due to a different metabolism or any other process, we only care it working.

With regards to the trade-off, we might lose some traits because they are not relevant for selection.

9.2 Metabolic engineering

Metabolic engineering is based on 4 general principles:



- **Gene overexpression:** Increasing the production of a gene's product by enhancing its transcription or translation.
- **Gene down-regulation:** Reducing gene activity to lower the amount of its product, often via RNA interference or repressors.

- **Gene deletion:** Completely removing a gene from the genome to stop its function.
- **Expression of heterologous genes:** Inserting and expressing a gene from another organism to introduce new functions.

Advantages	Disadvantages
Change of specific phenotypes, normally no screening is required	Genetic basis of desired phenotype needs to be known
Low risk of trade-offs	Genetic tools needed to be available
Traits are transferable across kingdoms	GMO classification can create limitations

